

Devarsh Vora

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Summary

Data & AI/ML Engineer with 3+ years spanning NLP, healthcare data, and production AI application development. Applies machine learning foundations to retrieval-augmented generation, document extraction, and agent evaluation. Experience includes Python APIs on AWS, Claude integration, and supervised fine-tuning using Llama and QLoRA. Translates business requirements into measurable evaluation criteria, investigates model limitations, and incorporates domain feedback into development. Brings a practical understanding of data quality, software testing, and tradeoffs between accuracy, latency, and inference cost.

Professional Experience

Data Scientist, CyberdomeUSA

08/2025 – Present | Indiana, USA

- Evaluated RAG faithfulness and context recall using RAGAS, tracking latency and token usage through Langfuse to assess quality and performance tradeoffs across retrieval and prompt configurations on fixed test cases before releases.
- Implemented document ingestion and hybrid retrieval with BM25, dense embeddings, and reranking, maintaining source versions and client access filters so product assistants retrieved current information from authorized knowledge bases.
- Developed document-extraction workflows for client applications using structured prompts and Pydantic validation, standardizing submission records while routing missing fields, conflicting evidence, and unsupported values for review.
- Containerized FastAPI services for AI product features and supported AWS deployment, adding request validation, timeouts, bounded retries, and GitHub Actions checks to release updates and restore previous configurations.
- Monitored the support assistant through Langfuse traces and client feedback, reviewing citation accuracy, unsupported answers, latency, and token usage to prioritize retrieval fixes and evaluate prompt changes before releases.

Applied AI Analyst (Contract), Matrices

09/2025 – 03/2026 | Remote, SF, USA

- Prepared versioned tasks for tool-using agents, specifying starting conditions, expected actions, and completion criteria, then reviewing ambiguous cases with project leads to improve evaluation consistency across repeated benchmark runs.
- Developed Python test workflows with LangGraph and MCP tools, validating arguments and approval handling in controlled environments while documenting failed actions and reproducible traces for engineers investigating agent behavior.
- Evaluated agent runs using record-state assertions and human scoring rubrics, identifying 8 distinct defects in tool selection, retries, and duplicate updates, then adding regression cases before subsequent releases.
- Compared model and prompt configurations on fixed agent tasks, analyzing completion, tool-use errors, latency, and token costs to document tradeoffs and recommend changes for engineering review before subsequent releases.

Graduate Assistant, The University of Texas at Arlington

03/2024 – 07/2025 | Texas, USA

- Prepared datasets from public and faculty-approved documents using Python, removing duplicates, organizing reference summaries, and maintaining document-level splits for repeatable and comparable supervised summarization experiments.
- Ran supervised Llama-3.1-8B-Instruct experiments using QLoRA and Hugging Face PEFT, recording checkpoints and configurations, then comparing summary factuality against prompted baselines on held-out documents for faculty review.
- Developed a local Streamlit prototype tested with 6 academic staff to compare generated summaries against source passages, collect factual corrections, and export feedback for faculty discussions and experiment refinement.

Data Scientist, CrystalVox Limited

12/2020 – 12/2022 | Gujarat, India

- Fine-tuned BERT classifiers for payer correspondence using PyTorch and Hugging Face, comparing results with TF-IDF baselines and routing uncertain predictions to billing specialists before incorporating reviewed labels into retraining.
- Integrated Tesseract OCR and spaCy extraction for billing documents, achieving 90% field-level exact-match accuracy on 200 held-out forms before correction, with validation checks for provider identifiers and service dates.
- Implemented semantic search over client documentation and resolved cases using Sentence-BERT and FAISS, separating client indexes and returning source references to help operations teams investigate recurring service requests.
- Packaged document classifiers into Dockerized FastAPI endpoints with request validation and versioned model artifacts, working with application developers to integrate predictions into client workflows and investigate failed processing requests.
- Maintained scheduled NLP processing jobs and reviewed specialist corrections, checking classification errors by client and document type before updating training datasets and comparing replacement models against the deployed baseline.

Technical Skills

- **Programming & APIs:** Python, SQL, Bash, pandas, NumPy, FastAPI, Pydantic, REST APIs, pytest
- **Machine Learning & NLP:** PyTorch, scikit-learn, Hugging Face Transformers, BERT, Sentence Transformers, spaCy, PEFT, LoRA/QLoRA, Fine-Tuning, Text Classification, Text Summarization, Named Entity Recognition, OCR, Model Evaluation
- **Generative AI & Agents:** Claude, Llama, RAG, Prompt Engineering, Structured Outputs, Tool Calling, LangChain, LangGraph, MCP, Context Management, Output Validation, LLM Evaluation, Regression Testing, Prompt Versioning, Human-in-the-Loop
- **Retrieval & Data:** Embeddings, Document Chunking, Hybrid Search, Reranking, Qdrant, FAISS, PostgreSQL, Data Validation
- **Cloud & LLMops:** AWS Bedrock, Docker, MLflow, Langfuse, RAGAS, Git, GitHub Actions, CI/CD, Model Serving, Batch Inference

Education

Master of Science, University of Texas at Arlington

08/2023 - 05/2025 | Texas, USA

Computer Science | Specialty: Data Science

Bachelor of Technology, Gujarat Technological University

07/2019 - 07/2023 | Gujarat, India

Information & Communication Technology | Specialty: AI/ML